

C2_02 Gate A Protected Review - Astra RC4

Marketplace seller performance. Open only after a saved independent Gate A attempt; then close the review and perform changed repair proof.

1. Objective controls

Control	Expected	Why
Orders	90	fact row count
Completed	45	status=completed
Completed value	86157.13	includes orphan seller order 88
Seller IDs	16 raw / 15 unique	seller_id 5 conflicts
Orphan seller	999 / order 88	must remain visible
Orphan event	order 9999	event join exception

2. Expected reasoning path

Start with grain: sellers_raw is master-key grain (but dirty); orders is one row/order; events is one row/event. Never sum order-level gross_value after an unconstrained event join.

Duplicate key: seller_id 5 has conflicting regions. The current Gate A order fixture happens not to use seller 5, so current order KPIs are not changed by that conflict. That does not make arbitrary dedupe safe; document and escalate the master-data conflict.

Orphan: order 88 / seller 999 is completed for 3253.11. If you INNER JOIN to sellers and lose it, completed value falls from 86157.13 to 82904.02. That is a control failure, not cleanup.

Decision grain: on trusted mapped rows, Silver/North has 14 completed of 22 eligible non-cancelled orders (63.64%), below Bronze/West and Gold/South (both 68.18%). State denominator and orphan treatment before recommending action; do not infer seller causality from this extract.

Time: build Apr-Jun typed month grain first, then LAG. Missing periods must be detected rather than treating the previous existing row as automatically previous month.

3. Critical failures

- Uses INNER JOIN to make orphans disappear without an explicit exception population.
- Uses DISTINCT as a generic fan-out repair instead of proving key/cardinality.
- Calculates a KPI at event grain while calling it ticket/order/invoice grain.
- Uses LAG before proving complete/deterministic time grain.
- Ranks a dirty dimension attribute before resolving a conflicting duplicate key.
- Reports an attractive total without independent row/distinct-key/money controls.
- Treats EXPLAIN or fast runtime as proof that business logic is correct.

4. Competency floors

Competency	Required proof
Grain/cardinality	Input/output grain; duplicate and orphan treatment; no unexplained fan-out.
SQL architecture	Readable CTE/subquery stages with controls at the correct grain.
Window/time	Partition/order/frame or LAG contract; deterministic period logic and gap awareness.
Validation	Rows, distinct keys, numerator/denominator and money independently reconcile.
Performance/handoff	Safe EXPLAIN evidence after correctness; query comments/assumptions and rerun path.

Changed repair proof: after review, close this PDF and solve a different grain/key/time case. Do not use the same Gate fixture as repair evidence. Gate B opens only when explicitly assigned.